

# Dongping Zhang

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## Research interests

My research studies how people evaluate and make decisions with AI systems, with a focus on trust, information reliability, and real-world impact. I combine behavioral science, machine learning, and experimental methods to analyze how users interpret model outputs, respond to uncertainty, and assess the trustworthiness of algorithmic recommendations. Through controlled experiments and empirical evaluation of deployed AI systems, I develop metrics and frameworks for evaluating AI outputs and their effects on user judgment and decision-making. My work focuses on advancing the responsible and reliable use of AI in high-stakes settings, including healthcare, digital platforms, and information-rich environments.

## Education

### Northwestern University

Evanston, IL

Ph.D., M.S. in Technology and Social Behavior

2018 - 2024

*Dual Ph.D. in Computer Science and Communication Studies*

Committee: Jessica Hullman (Chair), Jason Hartline, Matt Kay, Ágnes Horvát

Dissertation: *Strategies for Communicating Uncertainty in Predictive Systems for Enhanced Data-Driven Decision-Making*

### The University of Chicago

Chicago, IL

M.A. in Computational Social Science

2016 – 2018

Advisor: Luc Anselin

### University of California, Berkeley

Berkeley, CA

B.A. in Statistics, B.A. in Economics

2012 – 2016

## Peer-reviewed publications

**Zhang, Dongping**, Mukund Nakhate, Cheryl Stults, Heather Himelhoch, Anthony Solomonides, Sien Deng, Joseph Wilcox, Naveed Rabbani, Sachin Shah, Veena Jones, and Nirav Shah (2026). “Ambient AI Scribe Implementation and Patient Experience Across Three Health Systems: A Multi-site Observational Study.” Submitted to the *Journal of Medical Internet Research*, 2026. *Under review*.

**Zhang, Dongping**, Anthony Solomonides, Sonia Sultan, Rema Padman, and Nirav Shah (2026). “Benchmarking Accuracy and Reasoning Alignment in Proprietary Clinical LLM Systems: A Standardized Evaluation Study.” Submitted to the *American Medical Informatics Association Annual Symposium (AMIA)*, 2026. *Under review*.

**Zhang, Dongping**, Angelos Chatzimparmpas, Negar Kamali, and Jessica Hullman (2024). “Evaluating the Utility of Conformal Prediction Sets for AI-advised Image Labeling.” *Proc. 2024 CHI Conf. Hum. Factors Comput. Syst.*, Article 302. DOI: <https://doi.org/10.1145/3613904.3642446>

*Best Paper Honorable Mention (Top 5% of 4,028 paper submissions)*

**Zhang, Dongping**, Jason Hartline, and Jessica Hullman (2024). “Designing Shared Information Displays for Agents of Varying Strategic Sophistication.” *Proc. ACM Hum.-Comput. Interact.*, Volume 8, Issue CSCW1, Article 42, DOI: <https://doi.org/10.1145/3637319>

**Zhang, Dongping**, Eytan Adar, and Jessica Hullman (2021). “Visualizing Uncertainty in Probabilistic Graphs with Network Hypothetical Outcome Plots (NetHOPs).” *IEEE Trans. Vis. Comput. Graph.*, Volume 28, Issue 1, pp. 443-453. DOI: <https://doi.org/10.1109/TVCG.2021.3114679>

Open-sourced model

**Zhang, Dongping** and Uri Wilensky. “NetLogo Taxi Cabs Model”. *Center for Connected Learning and Computer-Based Modeling*, Northwestern University, Evanston, IL. <http://ccl.northwestern.edu/netlogo/models/TaxiCabs>

Industry  
research experience

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| <b>Research Scientist   Investigational Innovation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025 - present |
| <i>Endeavor Health Clinical Operations</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Evanston, IL   |
| PI: Nirav S. Shah, MD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |
| Role and Impact: Lead and execute independent research on AI-enabled clinical decision-support and healthcare systems within the Investigational Innovation team, the research arm of Endeavor Health’s innovation division. Design and conduct empirical studies evaluating the safety, quality, and implementation of AI-assisted technologies in clinical and operational settings. Develop and apply mixed-methods evaluation frameworks to assess AI system performance, interpretability, and clinical integration, producing peer-reviewed research and directly informing institutional AI governance and safety evaluation practices. Collaborate with clinical, informatics, quality, and data analytics stakeholders to identify research and funding opportunities, mentor junior researchers, and disseminate findings at national venues. |                |
| <b>Research Scientist   AI + Data Visualization</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2024 - 2025    |
| <i>National Renewable Energy Laboratory</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Golden, CO     |
| <i>(operated by Alliance for Sustainable Energy, LLC)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |
| PI: Kristi Potter and Julianne Mueller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |

Role and Impact: Led research at the intersection of AI and data visualization, developing novel visualization techniques to support AI-assisted decision-making in renewable energy applications. Designed actionable uncertainty visualizations for large-scale surrogate models, enabling policymakers to interpret and interact with ensemble simulation outputs more effectively. Investigated methods to visualize distribution shifts introduced by differential privacy mechanisms to improve the interpretability, privacy, and robustness of AI-driven decision-support workflows. Collaborated with U.S. Department of Energy stakeholders and academic partners to secure competitive funding, mentor junior researchers, and contribute to research direction within the Computational Science Center.

Academic  
research experience

**Research Assistant** 2019 – 2024

*MU Collective Research Lab*, Northwestern University

PI: Jessica Hullman

Research Focus: Uncertainty quantification and communication in ML/AI systems for human–AI decision support using experimental methods.

**Research Assistant** 2018 – 2019

*Science of Networks in Communities*, Northwestern University

PI: Noshir Contractor

Research Focus: Social network analysis and modeling of large-scale communication networks to study tie formation and organizational dynamics.

Academic Services

**Associate Editor**

IEEE Journal of Biomedical and Health Informatics IEEE JBHI

**Reviewer**

The Journal of Visualization and Interaction JoVI 2026

**Reviewer**

The Journal of Visualization and Interaction JoVI 2025

**Program Committee / Reviewer**

IEEE Workshop on Uncertainty Visualization IEEE VIS 2025

**Reviewer**

IEEE Transactions on Visualization and Computer Graphics IEEE TVCG

**Program Committee / Reviewer**

TVCG Journal Paper Track IEEE PacificVis 2025

**Program Committee / Reviewer**

IEEE Workshop on Uncertainty Visualization IEEE VIS 2024

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching experience     | <b>Teaching assistant, Department of Computer Science (Northwestern)</b><br>COMP_SCI 333: Interactive Information Visualization<br>Fall 2025                                           |
|                         | <b>Teaching assistant, Department of Computer Science (Northwestern)</b><br>COMP_SCI 333: Interactive Information Visualization<br>Fall 2023                                           |
|                         | <b>Teaching assistant, School of Communication (Northwestern)</b><br>COMM.ST 395: Rhetoric of Sports Marketing<br>Spring 2022                                                          |
| Invited talks           | <b>Benchmarking Accuracy and Reasoning Alignment in Proprietary Clinical LLM Systems: A Standardized Evaluation Study</b><br>AMIA 2026 Annual Symposium                                |
|                         | <b>Designing Information Displays for Multi-agent Strategic Settings</b><br>ACM CSCW 2024                                                                                              |
|                         | <b>Uncertainty Quantification for AI-Advised Decision-Making</b><br>ACM CHI 2024                                                                                                       |
|                         | <b>Conformal Prediction for Deep Learning Classifiers</b><br>A Symposium on Human+AI, The University of Chicago, October 2023                                                          |
|                         | <b>Visualizing Uncertainty Embedded in Probabilistic Graph Models</b><br>IEEE VIS 2021 Virtual                                                                                         |
| Honors and scholarships | <b>Predictive Extensions to ERGMs and Applications in Real-time Monitoring of Organizational Social Networks</b><br>Seventh International Workshop on Social Network Analysis (ARS'19) |
|                         | <b>Segal Design Institute Research Cluster Fellowship</b><br>Northwestern University<br>2020 – 2021<br>Selected as a research fellow to advance knowledge of design innovation.        |
|                         | <b>Computational Social Sciences Tuition Award</b><br>The University of Chicago<br>2016 – 2018<br>Received a merit-based tuition scholarship during my M.A. program.                   |
| Skills                  | <b>Regents' and Chancellor's Scholarship</b><br>University of California, Berkeley<br>2012 – 2016<br>The most prestigious scholarship awarded to the top 2% of undergraduates.         |
|                         | <b>Programming:</b> R, Python, SQL<br><b>Web-based Prototyping:</b> HTML, CSS, JavaScript                                                                                              |

**Developer Tools:** Node.js, Bootstrap, Webpack, React, Firebase, Git, Figma

**Information Visualization:** D3.js, ggplot2, igraph, Tableau

**Qualitative Methods:** Ethnography, Research Interview, Observational Study, Survey Design, Design of Experiment

**Quantitative Methods:** Social Network Analysis, Agent-based Modeling and Simulation, Bayesian Modeling, Game Theory, Information Design, Data and Predictive Analytics, Machine Learning, Artificial Neural Network